

Michael Marko Sesay

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Professional Profile

■ Researcher, statistician, data scientist, and academic with a strong interdisciplinary background in **research methodology, quantitative analysis, academic writing, scientific communication, statistics, mathematics, and data science**. Holds a **First Class BSc (Hons) in Statistics** and an **MSc in Mathematics (Data Science Option)** from the Pan African University Institute for Basic Sciences, Technology and Innovation (PAUSTI), Jomo Kenyatta University of Agriculture and Technology (JKUAT), Kenya.

Experienced in the design and implementation of quantitative and computational research, including **research problem formulation, literature review, development of research objectives and questions, methodological design, data collection and management, statistical analysis, interpretation of results, academic reporting, and manuscript preparation**. Possesses a strong understanding of the complete research process, from initial concept development to the presentation and dissemination of research findings.

Skilled in **academic and scientific writing**, including research proposals, literature reviews, theses and dissertations, technical reports, journal manuscripts, referencing, citation management, presentation of results, and scholarly communication. Strongly committed to **academic integrity, clarity of scientific argument, evidence-based reasoning, reproducibility, and responsible research practice**. Capable of supporting students in developing well-structured research projects and communicating their findings clearly and professionally.

Proficient in **Python, R, MATLAB, SPSS, LaTeX, Microsoft Office**, with experience in statistical modelling, data preprocessing, predictive analytics, time-series analysis, model evaluation, data visualisation, and reproducible scientific computing. This quantitative and computational background provides an additional strength for supporting students undertaking data-driven and empirical research.

Author and co-author of **peer-reviewed scientific publications**, with experience in interdisciplinary research, manuscript development, data analysis, research communication, and evidence-based problem solving. Brings together strong analytical ability, scientific writing competence, teaching and mentoring potential, teamwork, and a commitment to strengthening students' capacity in **research methods, academic writing, quantitative reasoning, and scholarly communication**.

Employment History

- 2024 – 2026 ■ **Researcher**, African Foundation for Cognitive and Frontier Technologies (AFCFT), Pan African University Institute for Basic Sciences, Technology and Innovation (PAUSTI), Jomo Kenyatta University of Agriculture and Technology (JKUAT), Kenya.
Conducted interdisciplinary research in artificial intelligence, machine learning, statistical modelling, deep learning, and time-series forecasting, with applications in energy systems and public health.
- 2023 – 2024 ■ **ICT4D Consultant**, Catholic Relief Services (CRS), Sierra Leone.
Supported digital data collection, monitoring and evaluation, information management, data quality assurance, and evidence-based decision-making for development programmes.

Employment History (continued)

- 2021 – 2022  **Consultant**, Informatics Consultancy Firm, Sierra Leone.
Designed and managed electronic data-collection tools and supported data management, statistical analysis, reporting, monitoring and evaluation, and technology-driven solutions for research and development projects.

Education

- 2024 – 2026  **Master of Science in Mathematics (Data Science Option)**, Pan African University Institute for Basic Sciences, Technology and Innovation (PAUSTI), hosted at Jomo Kenyatta University of Agriculture and Technology (JKUAT), Kenya.
Status: Completed.
Scholarship: Pan African University Scholarship.
CGPA: 3.77/4.00.
Thesis: *An Explainable Hybrid Transformer–BiLSTM Framework for Multivariate Household Energy Demand Forecasting with Weather Integration*.
- 2018 – 2022  **Bachelor of Science with Honours in Statistics, First Class**, Fourah Bay College, University of Sierra Leone, Sierra Leone.
CGPA: 3.56/4.00.

Research Publications

Peer-Reviewed Journal Articles

- 1 M. M. Sesay, “Leakage-aware benchmarking of temporal data preparation for hourly energy forecasting,” *Energy Conversion and Management: X*, 2026.  DOI: 10.1016/j.ecmx.2026.101791.
- 2 M. M. Sesay and A. Ngunyi, “Beyond point forecasting: Quantile cnn-lstm load forecasting with uncertainty-aware battery scheduling for residential energy management,” *International Journal of Electrical Power & Energy Systems*, p. 112 076, 2026.  DOI: 10.1016/j.ijepes.2026.112076.
- 3 M. M. Sesay, A. Ngunyi, and H. Imboga, “An explainable hybrid transformer–bilstm framework for multivariate household energy demand forecasting with weather integration,” *Energy Informatics*, vol. 9, p. 33, 2026.  DOI: 10.1186/s42162-026-00635-8.
- 4 M. M. Sesay, A. Ngunyi, and H. Imboga, “Probabilistic forecasting of monthly dengue cases using epidemiological and climate signals: A bilstm–negative binomial model versus mechanistic and count-model baselines,” *PLOS Global Public Health*, vol. 6, no. 3, pp. 1–29, 2026.  DOI: 10.1371/journal.pgph.0005404.
- 5 M. S. Wembo, B. Malenje, and M. M. Sesay, “A non-stationary nb–inar(1) model with time-varying beta–binomial thinning: Properties and rolling estimation,” *Commun. Math. Biol. Neurosci.*, 2026.  DOI: 10.28919/cmbn%2F9839.

Skills

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| Languages |  English. |
| Programming |  Python, R, MATLAB, $\text{\LaTeX}$. |
| Machine Learning |  TensorFlow/Keras, PyTorch, scikit-learn, NumPy, Pandas, statsmodels. |
| Data Science |  Statistical modelling, machine learning, deep learning, time-series analysis, probabilistic forecasting, uncertainty quantification, predictive analytics, and data visualization. |

Skills (continued)

Data Management	■ KoboToolbox, CommCare, Red Rose, digital survey systems, data cleaning, validation, quality assurance, and reporting.
Research Tools	■ Overleaf, Git, GitHub, Microsoft Office, Google Workspace.
Professional	■ Academic research, scientific writing, teaching, training, monitoring and evaluation, ICT4D, data-driven decision-making, and interdisciplinary research collaboration.

Miscellaneous Experience

Awards and Achievements

- 2024 – 2026 ■ **Pan African University Scholarship**, awarded for MSc studies at the Pan African University Institute for Basic Sciences, Technology and Innovation (PAUSTI), Jomo Kenyatta University of Agriculture and Technology, Kenya.
- 2022 ■ **First Class Honours**, Bachelor of Science with Honours in Statistics, Fourah Bay College, University of Sierra Leone.
- 2026 ■ **Research Publication Achievement**, author/co-author of peer-reviewed journal articles published in internationally recognised journals in energy, artificial intelligence, forecasting, and public health.

Certifications and Professional Training

- 2025 ■ **Climate Policy Analysis and Mentorship Program**, Green Finance Hub. Completed training in global climate policy, sustainable finance, climate governance, carbon markets, adaptation, and mitigation frameworks.
- **Digital Skills, Entrepreneurial Skills, and Financial Literacy Training**, African Union CIEFFA and ImpactHER. Completed online capacity-building training in digital skills, entrepreneurship, and financial literacy.

References

Prof. Antony Ngunyi

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MSc Research Supervisor

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